

Original article

A multicenter study on enteral autonomy outcome of pediatric intestinal failure patients from a middle-income country

Heitor P. Leite ^a, Rodrigo Vincenzi ^b, Carlos O. Kieling ^c, Paulo C. Koch Nogueira ^a, Roberta L. Longo ^b, Natalia C. Person ^b, Joao Seda Neto ^b, Estela C. Pavanelli ^b, Catiana M. Gritti ^b, Mariana J.B.M. Fonseca ^d, Maria Fernanda C. de Camargo ^d, Camila P. Genzani ^d, Eduardo F. Hatanaka ^d, Keilla M.C.B. Uchoa ^d, Simone M.R.M. Perentel ^d, Marina R. Adami ^e, Marilia R. Ceza ^e, Daltro L.A. Nunes ^e, Berenice L. Santos ^f, Liege L. Godoy ^f, Leticia Feldens ^f, Helena A.S. Goldani ^{f,*}

^a Department of Pediatrics, Universidade Federal de São Paulo, Center for Intestinal Rehabilitation, Hospital Samaritano de São Paulo, São Paulo, Brazil

^b Intestinal Rehabilitation and Transplant Center, Hospital Sírio-Libanês, São Paulo, Brazil

^c Pediatric Intestinal Rehabilitation Center, Pediatric Gastroenterology Unit, Hospital de Clínicas de Porto Alegre, Porto Alegre, Brazil

^d Center for Intestinal Rehabilitation, Hospital Samaritano de São Paulo, São Paulo, Brazil

^e Pediatric Intestinal Rehabilitation Program, Pediatric Gastroenterology Unit, Hospital de Clínicas de Porto Alegre, Porto Alegre, Brazil

^f Pediatric Intestinal Rehabilitation Program, Post-Graduate Program of Child and Adolescent Health, Universidade Federal do Rio Grande do Sul, Hospital de Clínicas de Porto Alegre, Porto Alegre, Brazil

ARTICLE INFO

Article history:

Received 1 October 2024

Accepted 12 January 2025

Keywords:

Intestinal failure

Short bowel syndrome

Multicenter study

Children

SUMMARY

Background & aims: To identify predictors of enteral autonomy and survival in pediatric intestinal failure patients followed up at three pediatric intestinal rehabilitation centers from a middle-income country. **Methods:** This retrospective multicenter cohort study evaluated patients with intestinal failure from three high-volume intestinal rehabilitation centers on long-term parenteral nutrition between 2014 and 2023. The primary outcome was status at the end of the follow-up: parenteral nutrition dependence, enteral autonomy, transplantation, or death. Secondary outcomes were complications resulting from treatment and current parenteral nutrition dependency index. Competing risk analysis, Cox regression, and a decision tree model were employed.

Results: The cohort comprised 207 patients with a median (interquartile range) age at admission of 6.2 (3.1; 12.9) months and a median follow-up of 29.4 (10.2; 49.6) months. Short bowel (85 %) and motility disorders (10.6 %) were the primary causes of intestinal failure. Cumulative incidence of 5-year for enteral autonomy and survival rates were 37 % and 88 %, respectively. Intestinal failure-associated liver disease was present in 24 % of patients on admission. Enteral autonomy was associated with remnant intestine length >40 cm (hazard ratio: 2.0; 95 % confidence interval: 1.7; 3.6); age at admission <6.2 months (hazard ratio: 1.8; 95 % confidence interval: 1.0; 3.3); and preserved ileocecal valve (hazard ratio: 3.4; 95 % confidence interval: 1.9; 6.0). The overall mortality rate was 7.7 %.

Conclusion: The overall survival rate was 92.3 % and the 5-year cumulative incidence of enteral autonomy was 37 %. The probability of achieving enteral autonomy was associated with early arrival at the intestinal rehabilitation center and favorable bowel anatomy.

© 2025 European Society for Clinical Nutrition and Metabolism. Published by Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

* Corresponding author. Rua Ramiro Barcelos 2350, Bairro Santa Cecilia, CEP 90035-903, Porto Alegre-RS, Brazil.

E-mail addresses: heitorpons@gmail.com (H.P. Leite), rodrigo.vincenzi@hsl.org.br (R. Vincenzi), ckieling@hcpa.edu.br (C.O. Kieling), pkochnogueira@gmail.com (P.C. Koch Nogueira), roberta.l.longo@gmail.com (R.L. Longo), nataliacperson@gmail.com (N.C. Person), joaoseda@gmail.com (J.S. Neto), estela.pavanelli@gmail.com (E.C. Pavanelli), catianagritti@gmail.com (C.M. Gritti), marijbarcia@gmail.com (M.J.B.M. Fonseca), fernanda.camargo.hemoped@gmail.com (M.F.C. Camargo), genzani@hotmail.com (C.P. Genzani), hatanaka@uol.com.br (E.F. Hatanaka), mayumi_uchoa@yahoo.com.br (K.M.C.B. Uchoa), simone.perentel@samaritano.com.br (S.M.R.M. Perentel), mradami@hcpa.edu.br (M.R. Adami), mrceza@hcpa.edu.br (M.R. Ceza), dlunes@hcpa.edu.br (D.L.A. Nunes), berlemppek@gmail.com (B.L. Santos), lgodoy@hcpa.edu.br (L.L. Godoy), lfeldens@hcpa.edu.br (L. Feldens), hgoldani@hcpa.edu.br (H.A.S. Goldani).

1. Introduction

Intestinal failure (IF) is defined as a reduction in functional intestinal mass below the minimum amount required for adequate digestion and absorption of nutrients and liquids for growth in children and weight maintenance in adults [1]. Short bowel syndrome (SBS) is characterized by a state of severe malabsorption, usually requiring parenteral nutrition (PN) for a period of more than 60 days after small bowel resection or intestinal length below 25 % of that expected for age [2]. SBS is the most common cause of IF and is generally caused by intestinal resection resulting from necrotizing enterocolitis, intestinal atresia, volvulus, gastroschisis, and Hirschsprung's disease [3].

Enteral nutrition, home parenteral nutrition (HPN), and clinical and surgical strategies to prevent or minimize complications [4,5], are the basis of treatment. This should be performed in multidisciplinary intestinal rehabilitation programs (MIRP) [2,6–8], whose main goals are to promote intestinal adaptation and enteral autonomy (EA). MIRPs are well established in Europe and North America, and the experiences of new centers have been described in developing countries [9–11]. Multicenter studies describing the course and clinical outcomes of cohorts of pediatric patients with IF in North America, the UK, New Zealand, France, and Scandinavia [12–14] have recently been published. However, there is a lack of longitudinal multicenter studies on the characteristics, outcomes, and prognostic factors of patients followed up at MIRPs in middle-income countries.

Although the treatment of children with IF is costly and complex, we hypothesized that the clinical practices carried out in middle-income countries may yield outcomes comparable to those of centers in North America, Europe and Oceania. Therefore, we aimed to evaluate the clinical outcomes of patients from three high-volume MIRPs in Brazil, a middle-income country in South America.

2. Material & methods

In this retrospective cohort study, we analyzed data extracted from the databases of patients diagnosed with IF between January 2014 and December 2023, followed up in three high-volume (more than 20 patients on home PN) MIRPs in the Southern and Southeastern regions of Brazil: Hospital de Clinicas de Porto Alegre, Hospital Menino Jesus/Hospital Sirio Libanes in São Paulo, and Hospital Samaritano de São Paulo. The first two programs are supported by the Ministry of Health and the latter provides care to private and health insurance patients. The study was approved by the Ethics and Research Committees of the three institutions. Written informed consent was obtained from the parents or legal guardians of the patients.

The three MIRPs have a multi-professional intestinal rehabilitation team made up of pediatricians, pediatric gastroenterologists, pediatric surgeons, intensivists, nutrition specialists, dietitians, specialized nurses, psychologists, pharmacists, physiotherapists, speech therapists and social workers. Each program provides formal training protocols for the parents and family members, as well as local health professionals from the patients' municipality of origin, regarding the care of the HPN process, central venous catheter (CVC), and infusion pumps, according to international guidelines [15].

2.1. Standard of care of the three MIRPs

2.1.1. Parenteral nutrition

PN solutions were individualized, supplied by formally contracted pharmaceutical companies, and delivered to patients'

homes as “3 in 1” bags containing amino acids, lipids, carbohydrates, electrolytes, and micronutrients. The energy requirements were calculated according to Schofield's formula [16] with an increase of 1.5–2 times the calorie requirements for patients who needed catch-up growth [17]. The SMOF lipid emulsion containing 30 % soybean oil, 30 % medium-chain triglycerides, 25 % olive oil, and 15 % fish oil was routinely used in the three centers. It was switched to a 2- to 8-week regimen of fish-oil-based lipid emulsion for patients with progressive cholestasis when conjugated bilirubin persistently exceeded 2 mg/dL.

2.1.2. Enteral and oral feeding

Oral feeding, including breastfeeding, was always encouraged, unless contraindicated. Enteral nutrition was administered via nasogastric tube using an amino acid-based or extensively hydrolyzed formula. For expected use beyond 2 months, percutaneous gastrostomy was employed.

2.1.3. Central venous access

Tunneled CVCs (Browiac, Hickman, Proline) were used in all patients. Peripherally Inserted Central Catheter (PICC) was used during the initial stage of hospitalization, particularly for newborns and young infants. All centers adopted a uniform protocol for the taurolidine-citrate lock solution (Taurollock®). CVC thrombosis was assessed semiannually or annually using Doppler ultrasound of the vessels or angiotomography, when necessary.

2.1.4. Training for home PN care and hospital discharge

Mothers and family members were formally trained by specialized nurses for no less than two weeks or 30-h period in HPN care [18]. Nurses from the patients' municipality of origin were trained prior to hospital discharge. Family members were supervised daily in the installation of PN and CVC care for 2–3 months; from then on, they could independently manage the process at the discretion of the local nurses, under 24-h on-call supervision by doctors and nurses.

2.1.5. Patient follow-up and weaning off PN

The start of patient follow-up in the study was at the date of the admission to the intestinal rehabilitation program. All patients had outpatient visits two weeks after discharge from the hospital and monthly, bimonthly, and four-monthly visits thereafter. Periodic evaluations included anthropometric parameters, liver and central venous vessels ultrasound, hematological evaluation, serum and urine electrolytes, nutritional status for vitamins and trace elements, lipid profile, TSH, PTH, and liver and kidney function tests. Enteral autonomy was defined as the discontinuation of PN, with the ability to sustain growth and fluid balance by means of enteral support without parenteral support for more than three consecutive months [19].

2.2. Inclusion criteria

The study included patients aged <18 years diagnosed with primary intestinal disease requiring PN for over 60 consecutive days. Data were collected from July 1, 2023 to December 31, 2023, covering the period from admission to the latest evaluation. Primary causes of IF considered were SBS, congenital enteropathies, neuromuscular intestinal disease, and inflammatory bowel disease. Patients needing PN for non-digestive tract-related causes such as immune deficiency, cancer, or lymphatic/metabolic disorders were excluded. None received gastrointestinal hormones such as glucagon-like peptide (GLP-2).

2.3. Study variables and definitions

The primary endpoint was patient status at the end of the study period: PN dependence, enteral autonomy, intestinal transplantation, or death. Secondary endpoints were complications resulting from treatment, anthropometric nutritional status, and patients' current PN dependency index (PNDI). The PNDI was considered as the proportion of non-protein energy supply by PN (kcal/kg/day) in relation to the basal metabolic rate estimated using the Schofield formula (kcal/kg/day), categorizing dependence from mild (<80 %) to high (>120 %) [20,21].

Potential predictive variables for EA outcome were: age, gestational age at birth, sex, age at admission in MIRP defined as a binary variable according to the median age of the entire sample as > or <6.2 months, cause of IF, intestinal failure-associated liver disease (IFALD), length of remnant small intestine (binary variable: greater or less than 40 cm), presence of ileocecal valve (ICV), and type of SBS according to the anatomy of the remaining intestine: type I – terminal jejunostomy, type II – remnant intestine with jejunocolon anastomosis, type III – remnant intestine with jejunoleocolon anastomosis and preservation of ICV [4]. For death outcome, these same variables were considered. Additionally, data on type of health assistance, discharge on HPN use, follow-up duration, and PN use were collected.

Central venous catheter-associated bloodstream infection (CRBSI) was defined as clinical evidence of a systemic infection and at least one positive blood culture from the CVC and/or peripheral vein in the absence of an obvious source of infection other than the CVC [22]. The annual incidence rate of CRBSI was calculated using the incidence density ratio, defined as the number of infections per 1000 days of catheter use.

Intestinal failure-associated liver disease (IFALD) was defined as direct bilirubin values ≥ 2 mg/dL (34.2 μ mol/L) for two consecutive weeks, without concurrent sepsis or other causes of liver injury, after PN exposure for a minimum of 2 weeks [23–25]. Conjugated bilirubin was chosen as a marker for IFALD as it represents an accurate measure of cholestatic liver disease [26].

Anthropometric nutritional status was assessed by the z-scores of weight-for-age (W/A), height-for-age (H/A), and body mass index-for-age (BMI/A), compared to the WHO reference standard [27] and calculated using the WHO *AnthroPlus* software version 1.0.4 (WHO, Geneva, Switzerland). Scores on admission to the MIRPs and those obtained at the most recent evaluation before December 2023 were used.

2.4. Statistical analysis

Categorical variables are expressed as frequency and percentage, and continuous variables are expressed as median and interquartile range (IQR) or mean and standard deviation (SD). A paired t-test was used to analyze differences in the means of the anthropometric z-scores at two points in time. We used competing risk analysis [28] to assess the cumulative incidence of EA, with death treated as a competing risk, and Cox regression modeling was employed to investigate the relationships between exposure variables and patient outcomes (EA and survival). Initially, each factor was individually analyzed for associations with outcomes (p -value <0.1).

Categorical variables are expressed as frequency and percentage, and continuous variables are expressed as median and interquartile range (IQR) or mean and standard deviation (SD). A paired t-test was used to analyze differences in the means of the anthropometric z-scores at two points in time. A survival curve was developed according to the Kaplan–Meier method, and Cox regression modeling was employed to investigate the relationships between exposure variables and patient survival. Missing data occurred randomly for the presence

of ICV variable ($n = 2$) and the short bowel anatomy variable ($n = 1$). Subsequently, we combined the preselected factors into a multivariable model by removing any non-significant variables using a backward stepwise approach. We verified that the proportional hazards assumption was met in all the Cox analyses. To prevent collinearity in the multivariable models, we calculated the variance inflation factors (VIFs) and removed the variables with high VIF values (>5) from the model.

A supervised machine-learning model based on a decision tree was developed to identify the key factors associated with EA. It is constructed through recursive data division with the aim of choosing attributes that provide the best data separation. The results are expressed as the *importance* of each explanatory variable, indicating the influence of a specific attribute on the model. In other words, *importance* indicates the degree of contribution of each attribute to the decision-making process of the decision tree. The “rpart” package was used to classify individuals, incorporating all pre-selected variables in the univariate Cox regression model. Finally, the decision tree was pruned to optimize the model's performance, and the optimal complexity parameter was determined by minimizing the error rate. This approach results in a more accurate and parsimonious decision tree. Intercooled Stata version 14.0 (Stata Corp LP, College Station, TX, USA) and R statistical software (version 4.3.3) in association with R Studio (version 2024.04.2 + 764, 2024 by Posit Software, PBC) were used for the statistical calculations. All statistical tests were two-tailed, and the significance level was set at $p < 0.05$.

3. Results

3.1. Patients' characteristics

A total of 207 patients were included in the analysis. All patients had complete data. The main characteristics of the patients are summarized in Table 1. The main primary causes of IF were SBS (85 %), motility disorders (10.6 %), and enteropathies (2.4 %). Ninety-nine patients (44 %) underwent intestinal reconstruction, with serial transverse enteroplasty performed in 20 patients, and tapering in 8 patients.

3.2. Outcomes

The main clinical outcomes of the patients during the follow-up period are shown in Table 2. A total of 178 (86 %) had been discharged on HPN. Twenty-nine (14 %) patients could not be discharged on HPN due to psychological or social impairments or because they did not reach a stable clinical condition. Of the 207 patients, 62 (30 %) achieved EA (33 after home PN and 29 after hospital PN) and 124 (59.9 %) remained PN-dependent. Two patients (1 %) underwent intestinal transplantation (one isolated intestine and one multivisceral), 16 (7.7 %) died, and 3 (1.4 %) lost follow up as they were transferred to another center.

The median (IQR) of PNDI at the latest follow-up was 88.9 (59.8; 110.3) %, showing a moderate dependence. Death was associated with sepsis in 8 patients, IFALD in 1, loss of venous access in 1, and other causes in 6 patients. Four patients (1.9 %) developed IFALD during the follow-up.

Anthropometric assessment on admission showed W/A and H/A below -2 z-scores in 65 % and 67 % of patients, respectively. There were significant increases in all anthropometric z-scores from admission to the last follow-up (paired t-test, $p < 0.001$) (Table 2).

3.3. Cumulative incidence of outcomes

The actual frequency of EA in the sample was 62/207 (30 %). In the competing risk analysis, the predicted cumulative incidence of

Table 1
Main characteristics of the patients.

Variable	Values
Age, current (months) ^a	41.5 (18.4; 75.4)
at Admission (months) ^a	6.2 (3.1; 12.9)
Gestational age at birth (weeks) ^b	34.7 (3.8)
Sex, n (%)	
Male	129 (62.3)
Main diagnoses, n (%)	
Intestinal atresia	51 (24.6)
Necrotizing enterocolitis	41 (19.8)
Gastroschisis	40 (19.3)
Volvulus	35 (16.9)
Berdon syndrome	6 (2.9)
Hirschsprung's disease	5 (2.4)
Intestinal pseudo-obstruction	5 (2.4)
Microvillus inclusion disease	3 (1.4)
Crohn's disease	2 (0.9)
Other	21 (10.1)
Type of SBS^c, n (%)	
I - terminal enterostomy	9 (4.3)
II - jejuno-colic	107 (51.7)
III - jejuno-ileo-colic	58 (28.0)
n/A	32 (16)
Length of remnant bowel (cm) ^a	38 (17; 80)
Colon in continuity, n (%)	105 (50.7)
Preserved ileocecal valve, n (%)	91 (44)
Health care system, n (%)	
Public	169 (81.6)
Private	38 (18.4)

^a Data are presented as median (IQR).^b Mean (SD).^c Classification in types I, II, or III based on the anatomy of the intestine after reconstructive surgery.**Table 2**
Clinical outcomes of the patients.

Variable	Values
Follow-up period (months)	29.4 (10.2; 49.6)
Duration of parenteral nutrition (months)	19.4 (7.1; 48.1)
Discharged on home parenteral nutrition, n (%)	178 (86)
(Anthropometric nutritional status*)	
Admission W/A z-score ^a	-2.71 (-4.63; -1.27)
Current W/A z-score ^b	-0.71 (-1.35; -0.14)
Admission H/A z-score ^c	-2.87 (-4.60; -1.70)
Current H/A z-score ^d	-1.35 (-2.33; -0.59)
Admission BMI/A z-score ^e	-1.64 (-3.20; -0.35)
Current BMI/A z-score ^f	0.27 (-0.51; 0.85)
IFALD upon admission, n (%)	51 (24.6)
IFALD during follow-up, n (%)	4 (1.9)
Overall CRBSI incidence ratio per 1000 catheter-days	0.73 (0.0; 3.3)
PNDI at the late follow up	88.9 (59.8; 110.3)
Achievement of enteral autonomy, n (%)	62/207 (30.0)
Age of enteral autonomy achievement (months)	17.7 (12.5; 25)
Intestinal transplantation, n (%)	2 (1.0)
Overall mortality rate, n (%)	16 (7.7)

Data are expressed as median (IQR) otherwise indicated. *Significant increases in weight-for-age (W/A) (a,b), height-for-age (H/A) (c,d) and body mass index-for-age (BMI/A) (e,f) z-scores during the follow-up period (*t* test for paired data, *p* < 0.001); IFALD: intestinal failure associated liver disease; CRBSI: catheter-related blood stream infection. PNDI: parenteral nutrition dependency index.

enteral autonomy increased over time, reaching 37 % at 5 years of follow-up, as depicted in Fig. 1.

The results of the multivariate Cox models for the probability of EA over 5 years are shown in Table 3.

Given the smaller proportion of patients followed up in the private hospital (33 children), a sensitivity analysis was performed using the competing risk survival regression technique on the 174 patients from public hospitals. In this subgroup, the cumulative incidence of enteral autonomy was 0.43, slightly better than that of

the full group. To explore this difference, we compared key variables between the public and private groups. Private hospital patients had a longer median follow-up (51.5 vs. 25.3 months) and a lower frequency of type III SBS (12 % vs. 31 %). There was also a trend toward older age at admission (7.2 vs. 6.2 months) and a lower frequency of preserved ICV (30.3 % vs. 46.6 %). This analysis and derived comparisons were included in the supplementary material (Supplementary Fig. 1 and Table 1).

According to our method, five potential predictors of EA were preselected to enter in the multivariable model, but the preserved ICV and SBS type exhibited collinearity (VIF = 6.6 for both variables); therefore, we developed the model excluding the SBS type variable. In the final model, three variables were significantly associated with a greater probability of EA; the results of this analysis are depicted in Table 3.

As a complementary analysis, we conducted a logistic regression model using the binary outcome for intestinal rehabilitation (yes/no) including the same exposure variables used in the Cox model. The results were similar to those obtained in the Cox model, which identified the same risk factors (Supplementary Fig. 2).

3.4. Decision tree and enteral autonomy

The decision tree model was developed using the five variables with *p* < 0.1 in the univariate Cox model. The results of this analysis revealed that the most relevant predictors of EA were SBS type (importance: 13), preserved ICV (importance: 10), and age at admission (importance: 6). This is illustrated in Fig. 2.

The area under the ROC curve (AUC) of the decision tree was 0.72 (95 % CI 0.63; 0.79), attesting an acceptable accuracy for the model. The absence of ICV in the figure is attributed to its high correlation with the SBS type. As a result, the decision tree algorithm selected only one of these variables to be included in the plot, because they provide similar information. This highlights the importance of considering the relationships between variables when interpreting the results of the decision tree analysis.

3.5. Survival

The actual frequency of deaths in the sample was 16/207 (7.7 %) and the predicted patients' survival probability was 88 % at five years of follow-up (Fig. 3).

Results of the univariate Cox model for the probability of survival revealed that only the occurrence of previous IFALD presented a tendency to lower survival over five years, as shown in Table 4.

4. Discussion

This is the first multicenter study to provide original data on the trends in EA and survival in a large cohort of patients from three high-volume MIRPs in Brazil, a middle-income country. We found that the cumulative incidence of EA over 5 years of follow-up was 37 %.

The proportion of patients who achieved EA in our study was lower than that reported in a recent multicenter study conducted in North America, Europe, and Australasia, which showed a 6-year cumulative incidence of EA of 53 % [13]. Moreover, a recent meta-analysis of the clinical outcomes of pediatric IF reported an EA overall rate of 54 % [29]. Recently, the International Intestinal Failure Registry introduced by the International Intestinal Rehabilitation and Transplant Association (IIRTA) showed a 5-year cumulative incidence of EA of 47 % in 362 IF pediatric patients followed up at 26 IR centers worldwide [25]. The improvement in EA rates over the last decades may be the result of multiple factors: advancements in neonatal care, the advent of MIRPs, the

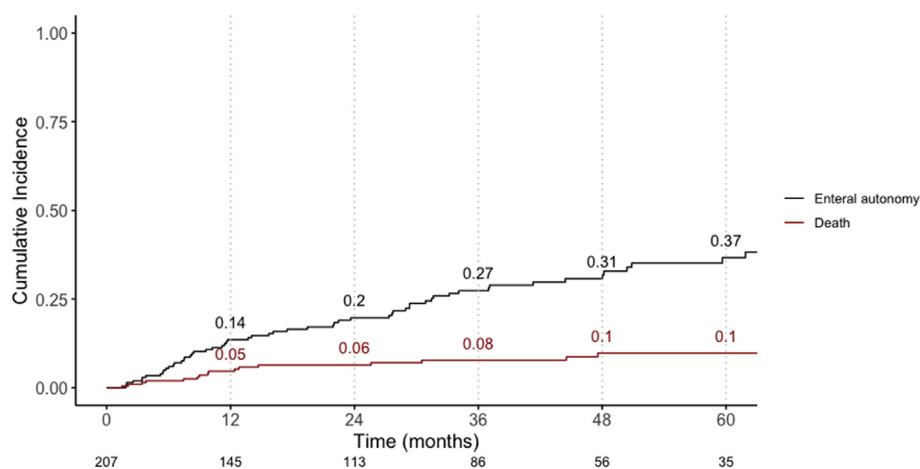


Fig. 1. Competing risk analysis of the probability of enteral autonomy over time, adjusted for death as a competing event.

Table 3
Multivariate Cox model of factors associated with enteral autonomy.

Variable	HR	95 % CI	p
Sex (male)	1.4	0.8; 2.4	0.196
Gestational age at birth (weeks)	0.9	0.9; 1.0	0.090
Preserved ileocecal valve	3.2	1.9; 5.5	<0.001
Remnant intestine (>40 cm)	2.6	1.5; 4.4	<0.001
Age at admission (<6.2 months)	1.9	1.1; 3.1	0.020
Primary disease			
intestinal Atresia	1	–	–
necrotizing Enterocolitis	0.8	0.4; 1.7	0.622
gastroschisis	1.2	0.6; 2.4	0.556
volvulus	0.5	0.2; 1.2	0.120
others	0.6	0.2; 1.3	0.187
IFALD (present)	0.7	0.4; 1.3	0.273
Type III SBS	5.7	3.1; 10.1	<0.001

HR: hazard ratio; CI: confidence interval; SBS: Short Bowel Syndrome.

development of new lipid emulsions, novel surgical approaches and techniques, and the reduction of septic episodes [30]. The scenario of IRPs in middle-income countries is highly challenging, primarily because of limited financial resources and insufficient access to scarce referral centers. Additional obstacles include lack of training for multidisciplinary teams, delays in diagnosis caused by low awareness, and logistical challenges in implementing home PN care [9]. In our study, age >6.2 months at admission to the MIRPs was negatively associated with EA. Other multicenter studies showed better EA outcomes when SBS diagnosis was made as early as 28 days of life [14], and worse outcomes (intestinal transplantation or death) when an IF diagnosis was made after 1 year of age [13].

Regarding the timing of referral to MIRPs, the severity of liver disease at the time of referral was associated with increased mortality in a sample of patients referred at a median age of 6.4 months, suggesting that early referral to MIRPs could improve liver disease outcomes [31]. It has been highly agreed that diagnosing and treating IF in specialized care centers at an early stage can significantly improve patient outcomes ([12,32,33]).

Late referral to MIRPs in our study may be due to limitations in the current patient referral flow of the public health system, as the two MIRPs supported by the Ministry of Health receive patients from long geographical distances across the country. Furthermore, 14 % of patients could not be discharged with home PN mainly due to familial and environmental factors that impaired home care. This may have contributed to the low EA observed in the study, despite

enteral autonomy sometimes being achievable during hospitalization when outpatient PN weaning is challenging [34].

In the multivariable Cox model used in our analysis, age at admission <6.2 months, remnant intestine >40 cm, and preserved ICV were associated with a greater probability of EA. These factors are part of a recent disease severity scoring tool developed to assess the probability of reaching EA: remnant small bowel >50 %, resection of the ICV, proportion of enteral nutrition, and conjugated bilirubin >34 $\mu\text{mol/L}$ [35].

Remnant intestine >40 cm and preserved ICV have been positively associated with EA in other multicenter studies [6,13,36,37]. Being treated at a non- or low-volume intestinal transplant (ITx) center was also a positive predictor of EA [6,13,36]. In our study, all MIRPs are non-ITx centers except for one. Preserving the ICV may represent the conservation of the proximal colon and distal ileum, which play crucial roles in intestinal adaptation, including their enteroendocrine functions and the highly specialized absorptive capacity of the ileum, as well as the absorption capacity of the proximal colon [30].

Complementing the multivariable Cox model analysis, the results of the decision tree reinforced the importance of the SBS type and age at admission in predicting EA. Decision trees are less affected by multicollinearity among predictor variables than regression-based models such as the Cox model. Since we selected all the pre-selected variables in the univariate Cox models to enter the decision tree, we consider that the two methods are complementary. The decision tree allowed us to demonstrate, in addition to the factors revealed in the Cox models, the central importance of the type of SBS, which could not be included in the regression model due to the presence of collinearity with the variable 'preserved ICV'. Finally, the decision tree pointed out that patients with IF are more likely to reach EA if they have SBS type III and start treatment at a MRP when <6.2 months of age (67 % probability). On the other hand, having SBS type I or II showed the lowest probability (17 %) of reaching EA.

The overall survival rate of 92.3 % found in our study was similar to that described in recent multicenter studies, which showed rates ranging from 91 % to 96.4 % [12–14,31]. Survival has been increasing [6], probably due to better organization of MIRPs and recent advances in managing patients with IF. Among the predictive factors evaluated in our study, only IFALD showed a trend of association with death. This is consistent with previous research, which has shown a positive relationship between IFALD and increased mortality [6,13,29,31].

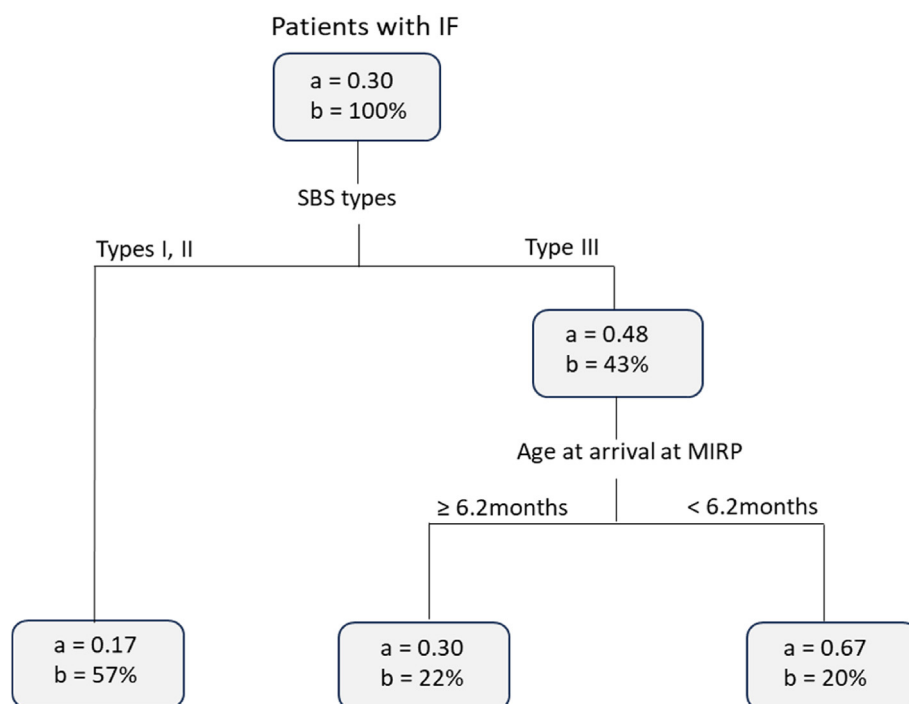


Fig. 2. Decision tree of factors associated with enteral autonomy (EA) in patients with intestinal failure (IF). The boxes (nodes) represent decision points based on a feature that contains: a) probability of the patients reaching EA; and b) percentage of all cases in the study sample reaching this node. SBS: short bowel syndrome; MIRP: Multidisciplinary Intestinal Rehabilitation Program.

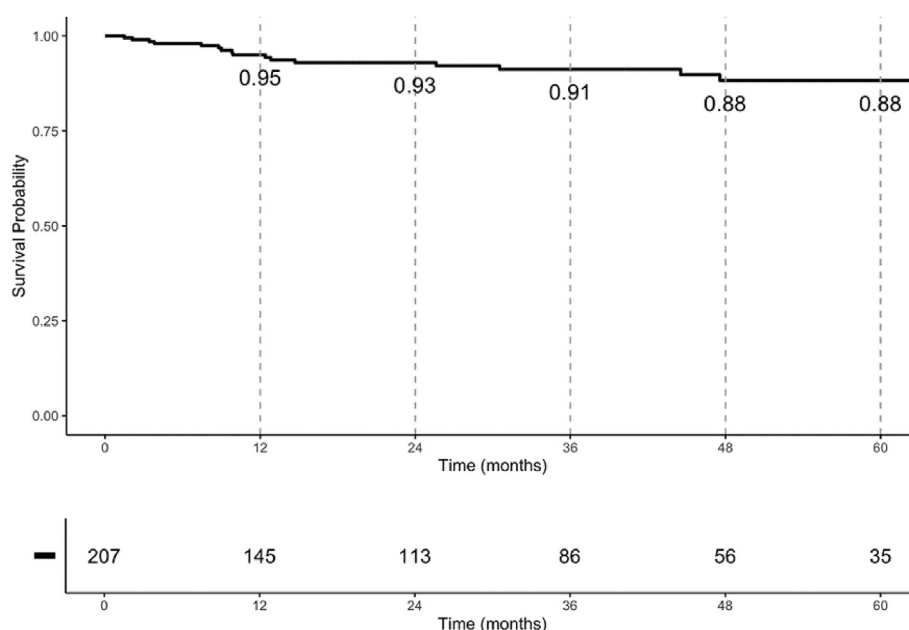


Fig. 3. Kaplan Meier's predicted probability of patients' survival over time.

IFALD was present in 24 % of our patients at admission, which is in line with other multicenter studies that reported an overall IFALD occurrence of 18 %–24 % [14,25]. During the follow-up, 4 patients developed IFALD. The low incidence of IFALD in our study may be attributed to several factors, such as early enteral nutrition, prevention and treatment of catheter-related sepsis, regular use of composite intravenous lipid emulsion (SMOFLipid®), and pure fish oil-based lipid emulsion when indicated. These management strategies are similar to those described in a French multicenter

study which reported an IFALD prevalence of less than 6 % [12]. However, it should be noted that the diagnosis of IFALD is not always precise, as the diagnostic criteria are not well established [19,24].

Other findings of our research showed a low CRBSI rate per 1000 catheter-days, which may be attributed to improvements in training protocols for family members and healthcare professionals involved with home PN, as well as the use of taurolidine locks [38]. The anthropometric parameters (W/A, H/A, and BMI/A z-scores)

Table 4
Univariate Cox model of factors associated with patient survival.

Variable	HR	95 % CI	p
Sex (male)	1.5	0.5; 4.3	0.451
Gestational age at birth (weeks)	0.9	0.8; 1.1	0.383
Preserved ileocecal valve	0.8	0.5; 3.2	0.712
Remnant intestine (>40 cm)	0.6	0.2; 2.0	0.418
Age at admission (<6.2 months)	0.5	0.2; 1.4	0.166
Primary disease			
Intestinal atresia	1	—	—
Necrotizing enterocolitis	0.6	0.1; 2.3	0.449
Gastroschisis	0.2	0.1; 1.9	0.176
Volvulus	0.2	0.1; 1.7	0.142
Others	1.0	0.3; 3.3	0.986
IFALD (present)	2.3	0.9; 6.3	0.093
Type III SBS	0.9	0.3; 3.1	0.923

HR: hazard ratio; CI: confidence interval; IFALD: intestinal failure-associated liver disease; SBS: Short Bowel Syndrome.

significantly increased from the entry point to the last assessment, demonstrating adequate parenteral and enteral nutrition, along with disease-specific treatment.

The limitations of this study are mainly related to its retrospective design and the lack of some recently published core outcomes, including central line-related thrombosis and health-related quality of life [39–42]. Although the three MIRPs have similar standardized protocols for patient treatment, there may be differences in management dynamics between private and public health settings where care is provided. Interestingly, a study showed that patients from different sociodemographic backgrounds had similar rates of EA [43]. In addition, a minimum follow-up length was not established. The rehabilitation programs are relatively new with cases being admitted recently. This sample represents the real-world situation in Brazil, which may be indicative of circumstances in other middle-income countries. Potential solutions for increasing the MIRP implementation in middle-income countries include advocating for government funding and public-private partnerships, investing in referral centers, training healthcare providers, and implementing educational campaigns to enable earlier diagnosis. Collaborative efforts to address these barriers may significantly improve patient outcomes and enhance the sustainability of IRPs in middle-income settings.

5. Conclusion

In conclusion, the probability of achieving EA was associated with early arrival at the MIRPs and favorable bowel anatomy. Our findings suggest that contemporary treatment at MIRPs enables pediatric IF patients in a middle-income country to achieve EA. The data underscore the importance of early referral to an IF specialized center to reduce complication rates and enhance EA outcomes.

Author contribution statement

HPL, COK, RV, PCKN, HASG participated in the conception and design of the study. RLL, NCP, JSN, ECP, CMG, MJBME, MFCC, EFH, CPG, KMCBU, SMRMP, MRA, MRC, DLAN, BLS, LLG, LF participated in the data collection. HPL, COK, PCKN, HASG participated in analysis and interpretation of data. HPL and HASG produced the first draft of the manuscript, and all authors were involved in producing the final version and approved its content.

Funding

This study was financed in part by Fundo de Incentivo à Pesquisa (FIPE) of Hospital de Clinicas de Porto Alegre.

Declaration of competing interest

The authors declare no conflict of interest

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.clnesp.2025.01.033>.

References

- [1] Duggan CP, Jaksic T. Pediatric intestinal failure. *N Engl J Med* 2017;377(7):666–75.
- [2] Merritt RJ, Cohran V, Raphael BP, Sentongo T, Volpert D, Warner BW, et al. Nutrition Committee of the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition. Intestinal rehabilitation programs in the management of pediatric intestinal failure and short bowel syndrome. *J Pediatr Gastroenterol Nutr* 2017;65(5):588–96.
- [3] Pironi L. Definitions of intestinal failure and the short bowel syndrome. *Best Pract Res Clin Gastroenterol* 2016;30(2):173–85.
- [4] Goulet O, Abi Nader E, Pigneur B, Lambe C. Short bowel syndrome as the leading cause of intestinal failure in early life: some insights into the management. *Pediatr Gastroenterol Hepatol Nutr* 2019;22(4):303–29.
- [5] Tappenden KA. Anatomical and physiological considerations in short bowel syndrome: emphasis on intestinal adaptation and the role of enterohormones. *Nutr Clin Pract* 2023;38(Suppl 1):S27–34.
- [6] Squires RH, Duggan C, Teitelbaum DH, Wales PW, Balint J, Venick R, et al. Pediatric Intestinal Failure Consortium. Natural history of pediatric intestinal failure: initial report from the Pediatric Intestinal Failure Consortium. *J Pediatr* 2012;161(4):723. 8.e2.
- [7] Avitzur Y, Wang JY, De Silva NT, Burghardt KM, Deangelis M, Grant D, et al. Impact of intestinal rehabilitation program and its innovative therapies on the outcome of intestinal transplant candidates. *J Pediatr Gastroenterol Nutr* 2015;61(1):18–23.
- [8] Goldani HAS, Ceza MR, Godoy LL, Giesta JM, Beier S, Oliveira JG, et al. Outcomes of the first 54 pediatric patients on long-term home parenteral nutrition from a single Brazilian center. *J Pediatr Gastroenterol Nutr* 2022;75(1):104–9.
- [9] Gondolesi GE, Pattin F, Nikkoupur H. Management of intestinal failure in middle-income countries, for children and adults. *Curr Opin Organ Transplant* 2018 Apr;23(2):212–8.
- [10] Spolidoro JVN, Souza MC, Goldani HAS, Tanzi MN, Busoni VB, Padilla M del C, et al. International Latin American Survey on pediatric intestinal failure team. *Nutrients* 2021;13(8):2754.
- [11] Rumbo C, Solar H, Ortega M, Busoni V, de Barrio S, Martinuzzi A, et al. Short bowel syndrome related intestinal failure outcomes in Latin America: insights from the RESTORE Registry. *JPEN - J Parenter Enter Nutr* 2024;48(8):956–64.
- [12] Goulet O, Breton A, Coste ME, Dubern B, Ecochard-Dugelay E, Guimber D, et al. Pediatric home parenteral nutrition in France: a six years national survey. *Clin Nutr* 2021;40(10):5278–87.
- [13] Gattini D, Roberts AJ, Wales PW, Beath SV, Evans HM, Hind J, et al. Trends in pediatric intestinal failure: a multicenter, multinational study. *J Pediatr* 2021;237:16–23.e4.
- [14] Mutanen A, Engstrand Lilja H, Wester T, Norrby H, Borg H, Persson S, et al. A nordic multicenter study on contemporary outcomes of pediatric short bowel syndrome in 208 patients. *Clin Nutr* 2023;42(7):1095–103.
- [15] Norsa L, Goulet O, Alberti D, Dekooning B, Domellöf M, Haiden N, et al. Nutrition and intestinal rehabilitation of children with short bowel syndrome: a position paper of the ESPGHAN Committee on Nutrition. Part 2: long-term follow-up on home parenteral nutrition. *J Pediatr Gastroenterol Nutr* 2023;77(2):298–314.
- [16] Joosten K, Embleton N, Yan W, Senterre T, ESPGHAN/ESPEN/ESPR/CSPEN working group on pediatric parenteral nutrition. ESPGHAN/ESPEN/ESPR/CSPEN guidelines on pediatric parenteral nutrition: energy. *Clin Nutr* 2018;37(6 Pt B):2309–14.
- [17] Abi Nader E, Lambe C, Talbotec C, Dong L, Pigneur B, Goulet O. A new concept to achieve optimal weight gain in malnourished infants on total parenteral nutrition. *JPEN - J Parenter Enter Nutr* 2018;42(1):78–86.
- [18] Hill S. Practical management of home parenteral nutrition in infancy. *Early Hum Dev* 2019;138:104876.
- [19] Modi BP, Galloway DP, Gura K, Nucci A, Plogsted S, Tucker A, et al. ASPEN definitions in pediatric intestinal failure. *JPEN - J Parenter Enter Nutr* 2022;46(1):42–59.
- [20] Abi Nader E, Lambe C, Talbotec C, Pigneur B, Lacaille F, Garnier-Lengliné H, et al. Outcome of home parenteral nutrition in 251 children over a 14-y period: report of a single center. *Am J Clin Nutr* 2016;103(5):1327–36.
- [21] Abi Nader E, Lambe C, Talbotec C, Acramel A, Pigneur B, Goulet O. Metabolic bone disease in children with intestinal failure is not associated with the level of parenteral nutrition dependency. *Clin Nutr* 2021;40(4):1974–82.
- [22] Wouters Y, Causevic E, Klek S, Groenewoud H, Wanten GJA. Use of catheter lock solutions in patients receiving home parenteral nutrition: a systematic

- review and individual-patient data meta-analysis. *JPEN - J Parenter Enter Nutr* 2020;44(7):1198–209.
- [23] Lee WS, Chew KS, Ng RT, Kasmi KE, Sokol RJ. Intestinal failure-associated liver disease (IFALD): insights into pathogenesis and advances in management. *Hepatol Int* 2020;14(3):305–16.
- [24] Gura KM, Elisofon SA. Intestinal failure-associated liver disease in infants - UpToDate [Internet]. [cited 2024 May 29]. Available from: <https://www.uptodate.com/contents/intestinal-failure-associated-liver-disease-in-infants>.
- [25] Avitzur Y, Pahl E, Venick R, the International Intestinal Failure Registry. The development of the international Intestinal Failure Registry and an overview of its results. *Eur J Pediatr Surg* 2024;34(2):172–81.
- [26] Javid PJ, Oron AP, Duggan CP, Squires RH, Horslen SP. Pediatric Intestinal Failure Consortium. The extent of intestinal failure-associated liver disease in patients referred for intestinal rehabilitation is associated with increased mortality: an analysis of the Pediatric Intestinal Failure Consortium database. *J Pediatr Surg* 2018;53(7):1399–402.
- [27] World Health Organization. (2006). WHO child growth standards: length/height-for-age, weight-for-age, weight-for-length, weight-for-height and body mass index-for-age. Methods and development. World Health Organization; 2006. Available from: <https://apps.who.int/iris/handle/10665/43413>.
- [28] Fine JP, Gray RJ. A proportional hazards model for the subdistribution of a competing risk. *J Am Stat Assoc* 1999;94(446):496–509.
- [29] Pierret ACS, Wilkinson JT, Zilbauer M, Mann JP. Clinical outcomes in pediatric intestinal failure: a meta-analysis and meta-regression. *Am J Clin Nutr* 2019;110(2):430–6.
- [30] Sandy NS, Roberts AJ, Wales PW, Toma RK, Belza C, Dogra H, et al. Small and large bowel anatomy is associated with enteral autonomy in infants with short bowel syndrome: a retrospective cohort study. *JPEN - J Parenter Enter Nutr* 2024;48(2):231–8.
- [31] Javid PJ, Malone FR, Bittner R, Healey PJ, Horslen SP. The optimal timing of referral to an intestinal failure program: the relationship between hyperbilirubinemia and mortality. *J Pediatr Surg* 2011;46(6):1052–6.
- [32] Premkumar MH, Soraisham A, Bagga N, Massieu LA, Maheshwari A. Nutritional management of short bowel syndrome. *Clin Perinatol* 2022;49(2):557–72.
- [33] Sheikh Z, Adigun A, Duro D. A retrospective review of the 5-year nutritional and clinical outcomes in pediatric patients undergoing intestinal rehabilitation. *Clin Nutr ESPEN* 2023;55:277–84.
- [34] Fialkowski A, Broyles K, DePaula B, Flores A, Gray M, Graham RJ, et al. Achieving enteral autonomy in children with intestinal failure following inpatient admission: a case series. *J Pediatr* 2024;275:114226.
- [35] Belza C, Fitzgerald K, de Silva N, Avitzur Y, Wales PW. Early predictors of enteral autonomy in pediatric intestinal failure resulting from short bowel syndrome: development of a disease severity scoring tool. *JPEN - J Parenter Enter Nutr* 2019;43(8):961–9.
- [36] Khan FA, Squires RH, Litman HJ, Balint J, Carter BA, Fisher JG, et al. Predictors of enteral autonomy in children with intestinal failure: a multicenter cohort study. *J Pediatr* 2015;167(1):29–34.e1.
- [37] Jiang W, Chen G, Wang Y, Zhong W, Zhou C, Zhang J, et al. Multi-center analysis of predictive factors of enteral autonomy and risk factors of complications of pediatric intestinal failure in China. *Front Pediatr* 2022;10:813865.
- [38] Vernon-Roberts A, Lopez RN, Frampton CM, Day AS. Meta-analysis of the efficacy of taurolidine in reducing catheter-related bloodstream infections for patients receiving parenteral nutrition. *JPEN - J Parenter Enter Nutr* 2022;46(7):1535–52.
- [39] Demirok A, Nagelkerke SCJ, Benninga MA, Köglmeier J, Mutanen A, Arnell H, et al. ERNICA Intestinal Failure Working Group and ESPGHAN Network of Intestinal Failure and Intestinal Transplant in Europe. Development of a core outcome set for pediatric chronic intestinal failure. *JPEN - J Parenter Enter Nutr* 2023;47(3):364–71.
- [40] Neam VC, Oron AP, Nair D, Edwards T, Horslen SP, Javid PJ. Factors associated with health-related quality of life in children with intestinal failure. *J Pediatr* 2020;216:13–18.e1.
- [41] Keefe G, Culbreath K, Staffa SJ, Carey AN, Jaksic T, Kumar R, et al. High rate of venous thromboembolism in severe pediatric intestinal failure. *J Pediatr* 2023;253:152–7.
- [42] Tran LC, Lazonby G, Morello R, Pham D, Ellis D, Goldthorpe J, et al. How good is quality-of-life for children receiving home parenteral nutrition? A pilot study. *Clin Nutr ESPEN*. 2019;29:119–24.
- [43] Enman MA, Wilkinson LT, Meloni KB, Shroyer MC, Jackson TF, Aban I, et al. Key determinants for achieving enteral autonomy and reduced parenteral nutrition exposure in pediatric intestinal failure. *JPEN - J Parenter Enter Nutr* 2020;44(7):1263–70.